Yük Dağılımı ve Yük Yoğunluğu

$C/m^3 = \rho$ 

$d = \frac{m}{V}$ (kütle yoğunluğu)

$\int dV = V$ hacmi verir (yoğunluğun V integrali)

$C/m^2 = \sigma$ 

Cismin yoğunluğu bazı yerlerde değişiyor ise, bu tür dağılıma **Sürekli Dağılım** denir.

dV

Dikdörtgenler prizması → $dV = dx \cdot dy \cdot dz$

İçi dolu silindir → $dV = 2\pi r\, dr\, dy$ (r yarıçap, y yükseklik)

Çizgisel yük Dağılımı (λ) → C/m

$Q = \int \lambda\, dl$ (l boyu)

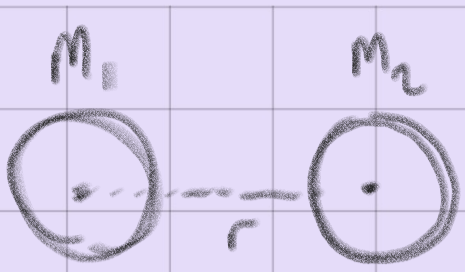
dl: Sonsuz küçük uzunluk

abcdefghijk
lmnopqrstuv
wxyz

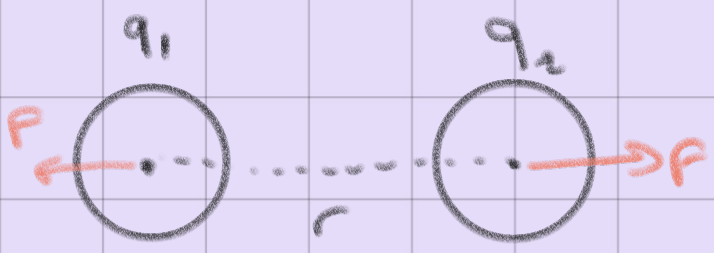
$\rho = Q/V$ $Q = \rho V$ $|Q_{p+}| = |Q_{e-}| = 1{,}6 \cdot 10^{-19}\ C$

$Q = \int \rho\, dV$ $Q, q \Rightarrow$ Elektrik yükü (skaler)

Çizgisel yük	Yüzeysel yük	Hacimsel yük
$\lambda = Q/L$	$\sigma = Q/A$	$\rho = Q/V$
$Q = \lambda \cdot L$	$Q = \sigma \cdot A$	$Q = \rho \cdot V$
$Q = \int \lambda \cdot dl$	$Q = \int \sigma \cdot dA$	$Q = \int \rho \cdot dV$
$dQ =$		



$$F_G = -G\frac{M_1 \cdot M_2}{r^2}$$ kütle çekimi



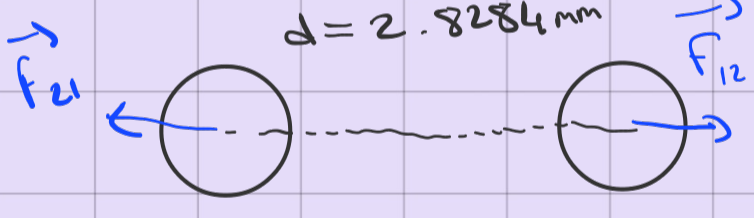
$$|F| = k\frac{|q_1 \cdot q_2|}{r^2}$$

$$k = 9 \cdot 10^9 \frac{N \cdot m^2}{C^2} = \frac{1}{4 \cdot \pi \cdot \varepsilon_0}$$

ε_0: Boş uzayın elektrik sabiti:

$$\varepsilon_0 \cong 8{,}85 \cdot 10^{-12} \frac{C^2}{N \cdot m^2}$$

Soru



Hareket başladıkları anda m_1 kütlenin ivmesinin büyüklüğü $a_1 = 9\ m/s^2$ diyor, m_2 kütleli parçacığın ivmesi $a_2 = ?$ ve $q = ?$

$$|\vec{F}_{12}| = |\vec{F}_{21}| = \frac{1}{4\pi\varepsilon_0} \cdot \frac{|q_1 \cdot q_2|}{d^2} = \frac{q^2}{4\pi\varepsilon_0 d^2}$$

$$\vec{F} = m \cdot \vec{a}$$

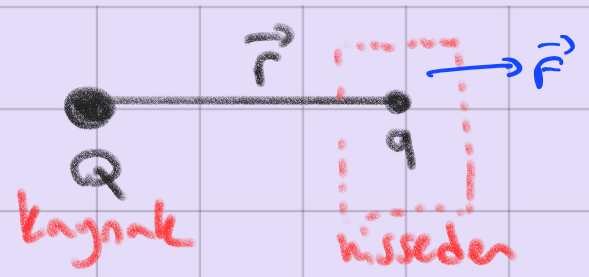
$$m_1 \cdot a_1 = m_2 \cdot a_2$$

$$a_2 = \frac{m_1}{m_2} \cdot a_1 = \frac{8 \cdot 10^{-7}}{6 \cdot 10^{-7}} \cdot 9 = 12\ m/s^2$$

$$F_{21} = \underbrace{8 \cdot 10^{-7}}_{m_1} \cdot \underbrace{9}_{a_1} = 72 \cdot 10^{-7}\ N$$

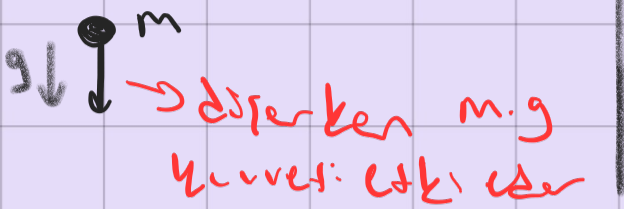
$$72 \cdot 10^{-7} = \underbrace{9 \cdot 10^9}_{k} \frac{q^2}{(2{,}8284 \cdot \underbrace{10^{-3}}_{\text{metre çevirme}})^2}$$

$$q = 8 \cdot 10^{-11}\ C = 80 \cdot 10^{-12}\ \text{piko coulomb}$$



$$\vec{F} = k \cdot \frac{Q \cdot q}{r^2} \cdot \hat{r}$$

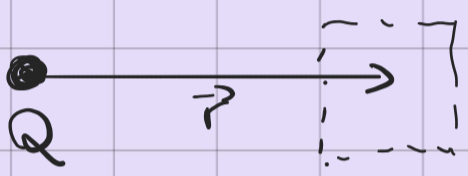
$$\hat{r} = \frac{\vec{r}}{r} \qquad \hat{r} : Q \rightarrow q$$



→ düşerken m.g kuvveti etki eder

g ↓ ortada m diye bişey yoksa sadece g

işte bunun gibi:



ortada bişey yoksa (q) → Elektrik alan vardır. Kuvvet yok

$$\vec{E}_Q(\vec{r}) = \frac{\vec{F}}{q} = k \cdot \frac{Q \cdot \cancel{q}}{r^2} \hat{r}$$

noktasal yükler için: $\vec{F}_q = q \cdot \vec{E}$

Ser buradan başla

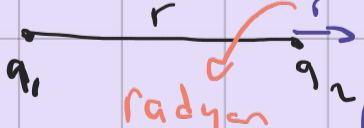
protonunki de bunun + hali $+1{,}6 \times 10^{-19}\,C$

e (elektron yükü) $= -1{,}6 \times 10^{-19}\,C$ dünyadaki en küçük yük miktarı

$k = 8{,}98 \times 10^9 \frac{N \cdot m^2}{C^2} = \frac{1}{4\pi\varepsilon_0}$ → boş uzayın elektriksel geçirgenlik katsayısı

Coulomb yasası

$$\varepsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N \cdot m^2}$$



radyan olarak dışarı doğru

$$F_C = \frac{q_1 \cdot q_2}{r^2} \cdot k$$

$$\boxed{\vec{F}_C = k \cdot \frac{q_1 \cdot q_2}{r^2} \cdot \hat{r}}$$

$|\hat{r}| = 1$ (birim vektördür. Sadece yön belirtir)

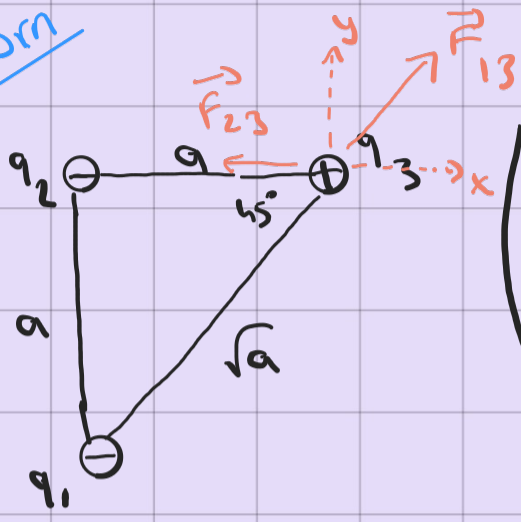
elektrostatik kuvvet = coulomb kuvveti

$|\vec{F_G}| = G \cdot \frac{m_1 \cdot m_2}{r^2}$ → kütle çekim kuvveti

$G = 6,67 \cdot 10^{-11} \frac{Nm^2}{kg^2}$

1 Coulomb = 10^{19} tane elektron yükü

1. Örn



mikro Coulomb

$q_1 = q_3 = 5\,\mu C$

$q_2 = -2\,\mu C$

$a = 0,1\,m$

$|\vec{F_{23}}| = (8,98 \times 10^9) \cdot \frac{(2\times10^{-6})(5\times10^{-6}\,C)}{(0.1)^2} = 8.99\,N$

$|\vec{F_{13}}| = k \cdot \frac{q_1 \cdot q_3}{(\sqrt{2}a)^2} = 11,2\,N$

$F_{13x} = F_{13} \cdot \cos 45 = 7.94\,N$

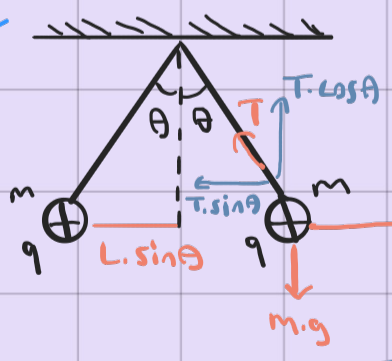
$F_{13y} = F_{13} \cdot \sin 45 = 7.94\,N$

$\vec{F_3} = F_{3x}\hat{i} + F_{3y}\hat{j}$

$\vec{F_3} = (7.94 - 8.99)\hat{i} + 7.94\hat{j}$

$\vec{F_3} = (-1.04\hat{i} + 7.94\hat{j})\,N$

2. Örn



$m = 3\times10^{-2}\,kg$

$L = 0.15\,m$

$\theta = 5°$

q ve T kaçtır?

$T\sin\theta = k \cdot \frac{q^2}{(2L\sin\theta)^2}$

$T\cos\theta = m \cdot g$

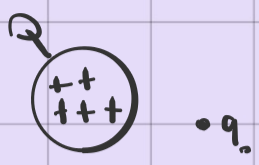
$\tan\theta = \frac{k \cdot q^2}{m \cdot g (2L \cdot \sin\theta)^2}$

$T = 3 \cdot 10^{-2} \cdot \frac{9,81}{\cos 5}$

$= 0,295\ldots$

$q = 4.42 \times 10^{-8}\,C$

Elektrik Alan



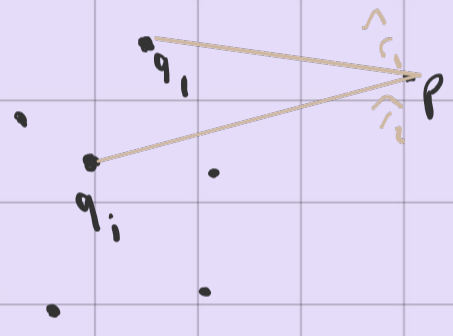
$$\vec{E} \equiv \frac{\vec{F_q}}{q_0} \left(\frac{N}{C}\right) \rightarrow \vec{F} = q \cdot \vec{E}$$

elektrik kuvvetin olması için cismin yüklü olması gerekir. yüksüz cisimler kuvvet etki etmez.

Noktasal Yükün Oluşturduğu Elektrik Alan



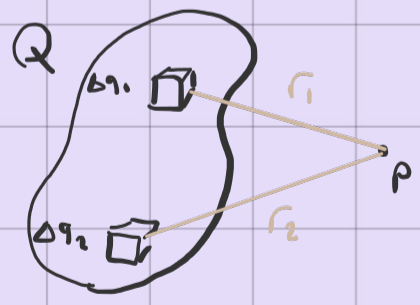
$$E_p = k \cdot \frac{q \cdot \cancel{1}}{r^2 \cdot \cancel{1}} \cdot \hat{r} = \boxed{k \cdot \frac{q}{r^2} \cdot \hat{r}}$$



$$\vec{E_p} = \vec{E_1} + \vec{E_2} + \cdots + \vec{E_i}$$

$$E_p = k \cdot \frac{q_1}{r_1^2} \hat{r}_1 + k \frac{q_2}{r_2^2} \hat{r}_2 + \cdots$$

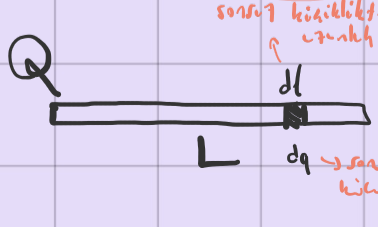
$$\boxed{E_p = k \sum_{i=1}^{\infty} \frac{q_i}{r_i^2} \hat{r}_i}$$



$$\vec{E_p} \approx k \cdot \frac{\Delta q_1}{r_1^2} \hat{r}_1 + k \frac{\Delta q_2}{r_2^2} \hat{r}_2 + \cdots$$

$$\approx k \sum_{i=1}^{\infty} \frac{\Delta q_i}{r_i^2} \hat{r}_i$$

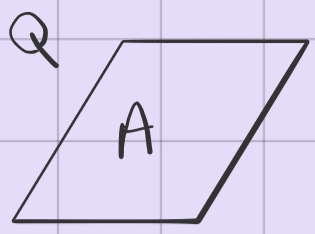
$$\vec{E_p} = k \cdot \lim_{\Delta q_i \to 0} \sum_{i=1}^{\infty} \frac{\Delta q_i}{r_i^2} \hat{r}_i \rightarrow \boxed{\vec{E_p} = k \int \frac{dq}{r^2} \hat{r}}$$



$$\lambda \equiv \frac{Q}{L} \left(\frac{C}{m}\right)$$

(Q: toplam yük, L: uzunluk)

$$\frac{\frac{L}{dl} \times \frac{Q}{dq}}{dq = \frac{Q}{L} dl \Rightarrow \boxed{dq = \lambda \, dl}}$$



sigma

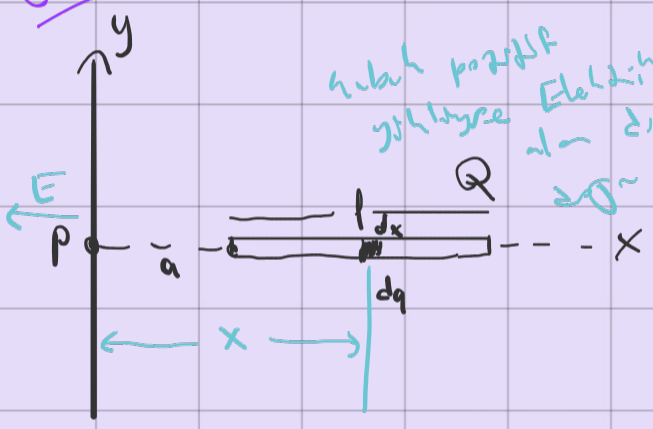
$$\sigma \equiv \frac{Q}{A} \left(\frac{C}{m^2}\right) \qquad \boxed{dq = \sigma\, dA}$$



ro

$$\rho \equiv \frac{Q}{V} \left(\frac{C}{m^3}\right) \qquad \boxed{dq = \rho\, dV}$$

Örn



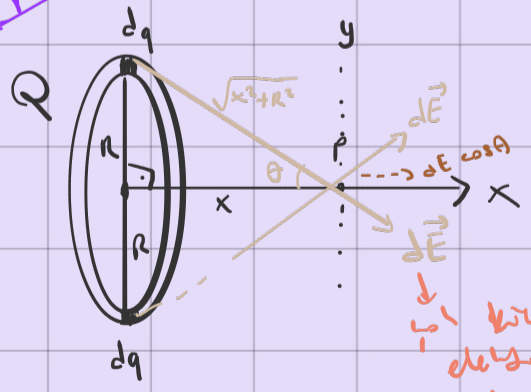
Sabit pozitif yüklüyse Elektrik alan dışa doğru olacak

$$E = k \int \frac{dq}{x^2}$$

$$E = k\lambda \int_a^{a+l} \frac{dx}{x^2} = k\lambda \left(\frac{1}{x}\Big|_a^{a+l}\right)$$

$$\vec{E} = -k\lambda \left(\frac{1}{\underset{(a)}{a+l}} - \frac{1}{\underset{(a+l)}{a}}\right) = \frac{-k \cdot \lambda \cdot l}{a(a+l)}\,\hat{\imath}$$

Örn



* Dikey bileşenler birbirini götürür.

Bir halkanın merkezi dışındaki herhangi bir noktada oluşturduğu elektrik alan sadece x eksenindedir, Sinüs bileşenleri birbirini yok eder.

Kosinüs bileşenleri kalır.

$$E_p = \int dE \cos\theta$$

$$E = k \int \frac{dq}{r^2} \cos\theta$$

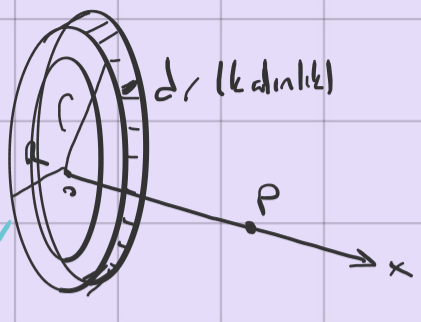
sabit değerlerdir, dışarı çıkarılır

$$E_p = k \int \frac{dq\, x}{(x^2+R^2)^{3/2}} = \frac{kx}{(x^2+R^2)^{3/2}} \underbrace{\int dq}_{Q}$$

$$dq = \lambda \cdot dl$$

$$\boxed{E_p = \frac{kQx}{(x^2+R^2)^{3/2}}}$$

Dişin Elektrik Alanı



diski küçük halkalara bölerek bulacağız

$$\sigma = \frac{Q}{A} = \frac{Q}{\pi \cdot R^2}$$

$$dq = \sigma \, dA$$

$$dA = 2\pi r \, dr$$

$$dq = \sigma \, 2\pi r \, dr$$

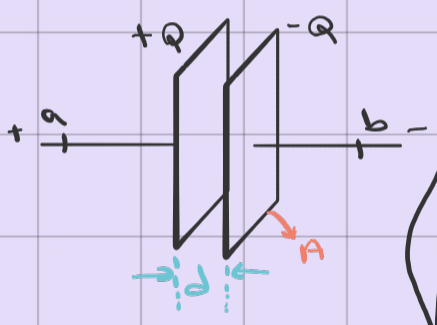
Sığaçlar (Kondansatör, Kapasitör)



$$V_{ab} = V_a - V_b$$

$Q = C \cdot V$ → volt

depo edilen yük ← Q (coulomb) — C: sığaç (Farad)



dolayısıyla

$E = \frac{\sigma}{\varepsilon_0}$ → birim yüzeye düşen yoğunluğu

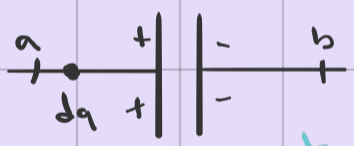
$\sigma = \frac{Q}{A}$

$E = \frac{V_{ab}}{d}$

$$\frac{V_{ab}}{d} = \frac{Q}{A \cdot \varepsilon_0}$$

$$\frac{Q}{V_{ab}} = \frac{\varepsilon_0 \cdot A}{d} = C$$

$$\boxed{C = \varepsilon_0 \cdot \frac{A}{d}}$$



da yükünü değiştirmeden yapılan iş

$$dW = dq \cdot V_{ab} = dq \, \frac{q}{C}$$

$$W = \frac{1}{C} \int_0^Q q \cdot dq = \frac{1}{2} \left(\frac{q^2}{2} \Big|_0^Q \right) = \boxed{\frac{Q^2}{2C}}$$

yerine

$Q = CV$

$$W = \boxed{\frac{1}{2} C V^2} = \boxed{\frac{1}{2} Q V}$$



$$h_{acim} = A \cdot d$$

$$u = \frac{1}{2} C V^2$$

ortam enerji yoğunluğu → $u = \frac{\frac{1}{2} C V^2}{A \cdot d} = \frac{1}{2} \cdot \frac{\varepsilon_0 A}{A d^2} (Ed)^2$

$$\boxed{u = \frac{1}{2} \varepsilon_0 \cdot E^2} \; J/m^3$$

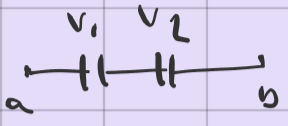
enerji yoğunluğu

(C sabit, Q VE V'ye bağlı değişiyor)

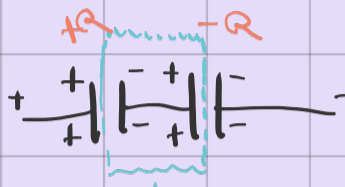
Sığaçların bağlanması

Seri

* Sığaçlar iletkendir.

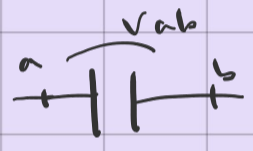


$V_{ab} = V_1 + V_2$



$Q = Q_1 = Q_2$

net yük sıfırdır



$C_{eş}$

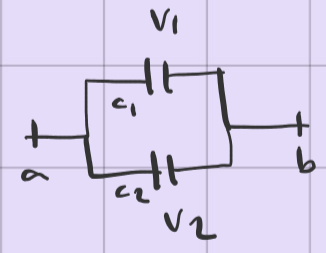
$Q = CV$

$$\frac{Q}{C_{eş}} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2}$$

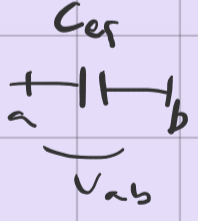
$$\boxed{\frac{1}{C_{eş}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots}$$

Sığaçlar seri bağlıysa kapasitörlerin terslerinin toplamı gibi toplanır.

Paralel



$V_{ab} = V_1 = V_2$

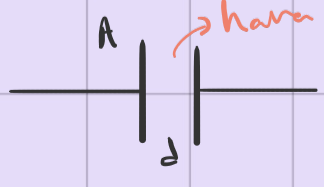


$Q = Q_1 + Q_2$

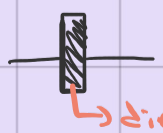
$C_{eş} \cdot V_{ab} = C_1 V_1 + C_2 V_2$

$$\boxed{C_{eq} = C_1 + C_2 + \cdots}$$

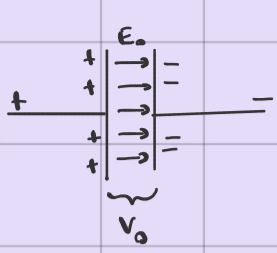
Sığaçlar ve Yalıtkanlar (Dielektrikler)



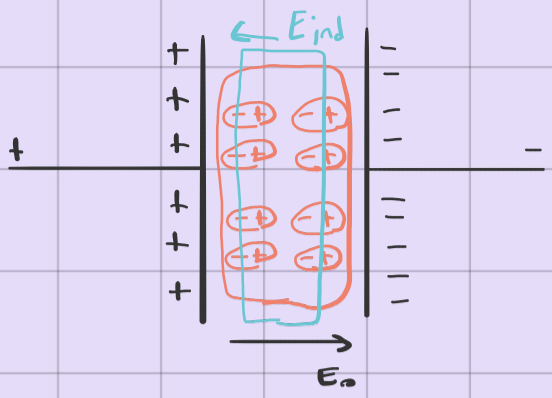
$$C_0 = \varepsilon_0 \frac{A}{d}$$



$$C = k \cdot C_0 = k \cdot \varepsilon_0 \cdot \frac{A}{d} \quad (k>1)$$



$$E_0 = \frac{V_0}{d}$$



$$E = E_0 - E_{ind} = \frac{E_0}{k}$$

$$V = \frac{V_0}{k}$$

$I = \frac{dQ}{dt}$ (coulomb / saniye) saniyede geçen yük yok

Kondansatörün yüklenmesi sırasında potansiyel fark ve akım şd. değişimi

$$V_{ab} = V_0 \left(1 - e^{-t/(Rc)}\right)$$

$$I = I_0 e^{-t/(Rc)}$$

$$q(t) = q_0 \cdot e^{-t/Rc}$$

$t = 0$ anında $q(0) = q_0 \cdot e^0 = q_0$

$t = \infty$ $q_{son} = q_0 \cdot e^{\frac{-\infty}{Rc}} = q_0 \cdot \frac{1}{e^{\infty}} = q_0 \cdot \frac{1}{\infty}$

$\frac{1}{\infty} = 0$ // boşalmıştır.

Manyetik Kuvvetler

yüklü bir parçacığa etki eden manyetik kuvvet:

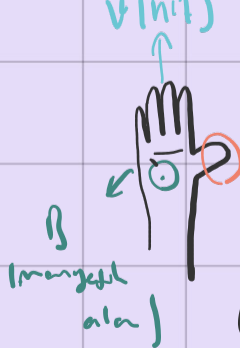
$$\vec{F} = q(\vec{v} \times \vec{B})$$

$\vec{F}$: N, q: C, $\vec{v}$: m/s, $\vec{B}$: T (manyetik alan, Tesla)

$F = |q| \cdot v \cdot B \cdot \sin\phi$ (Büyüklük)

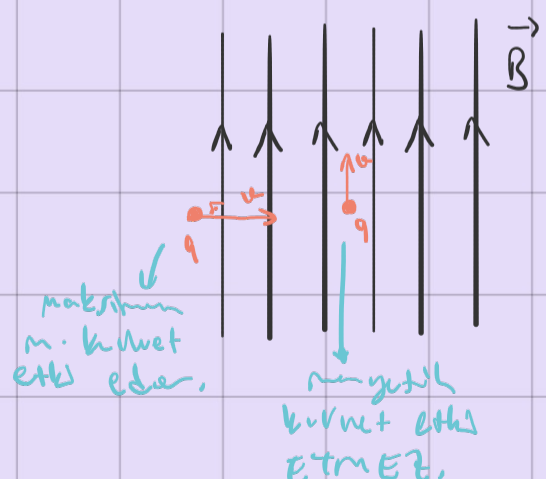
$\vec{B}$ ve $q\vec{v}$ → birleşimi

$\sin\phi$



(yük pozitif ise.)

(negatif ise kuvvetin yönü, bulunan yönün tersidir.)



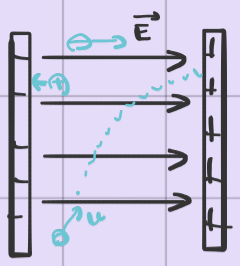
(birimden)

$$[B] = \frac{N}{C \cdot \frac{m}{s}} \text{ yada } \frac{N \cdot s}{C \cdot m} \equiv \text{Tesla (T)}$$

B'nin birimi

Hesap makinesinde shift + sin = $\sin^{-1}$ yani arcsin'i verir.

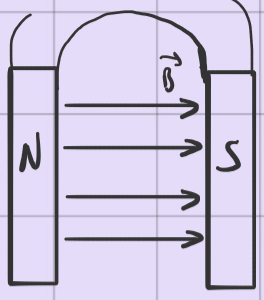
Elektrik kuv.



- yükü da yer
- kinetik enerjisi değişmez. yük iş yapar. M.a. ve hızı değişmez.

$W = \Delta k$

Manyetik kuv.



- Hareketsiz yükü ve
- Sadece yönü değiştirir. Dolayısıyla iş yapmaz.

$W = 0$

$\Delta k = 0$

1 proton:	$1,6 \times 10^{-19} C$
1 elektron:	$-1,6 \times 10^{-19} C$

Soru

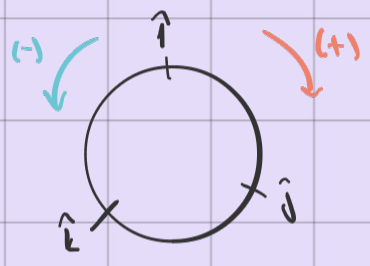
$$\vec{v} \times \vec{B} = (2\hat{i} - 4\hat{j} + \hat{k}) \times (\hat{i} + 2\hat{j} - \hat{k})$$

sadır

$$= (4\hat{k} + 2\hat{j}) + (4\hat{k} + 4\hat{i}) + (\hat{j} - 2\hat{i})$$

$$= 2\hat{i} + 3\hat{j} + 8\hat{k}$$

$$|\vec{v} \times \vec{B}| = \sqrt{(2)^2 + (3)^2 + (8)^2} = \boxed{8,77 \ \frac{m \cdot T}{s}}$$



Cismik Manyetik Alandaki Hareket

$$F_B = qvB = m \cdot a_r$$

merkezcil ivme

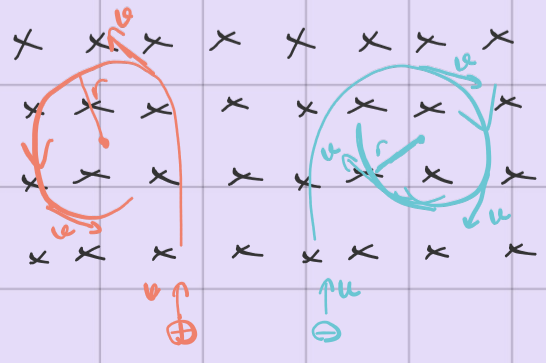
$$qvB = m \cdot \frac{v^2}{r}$$

$$\boxed{r = \frac{mv}{qB}}$$ → cismin kütlesi

açısal hız

$$w = \frac{v}{r} = 2\pi f$$

$$\boxed{w = \frac{Bq}{m}}$$ cismin açısal hızı


v
r
v
v
v

Elektrik alan + manyetik Alanın uyguladığı toplam kuvvet = **Lorentz kuvveti**

$$\vec{F_L} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\vec{F_L} = q[\vec{E} + (\vec{v} \times \vec{B})]$$

Hız Eleği

$F_E = F_B$ → Sadece bu değere sahip parçacıklar elekten geçer

$$qE = qvB$$

$$\boxed{v = \frac{E}{B}}$$

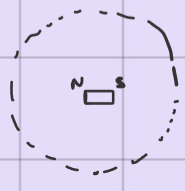
Elektrik Alan için Gauss Yasası

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{net}}{\varepsilon_0}$$



Manyetik Alan için Gauss Yasası

$$\oint \vec{B} \cdot d\vec{A} = 0$$



Manyetik Akı $\Phi = \vec{B} \cdot \vec{A}$

Akım taşıyan tele etki eden manyetik k.

$$F = BIl$$

$$\vec{F} = I(\vec{l} \times \vec{B})$$